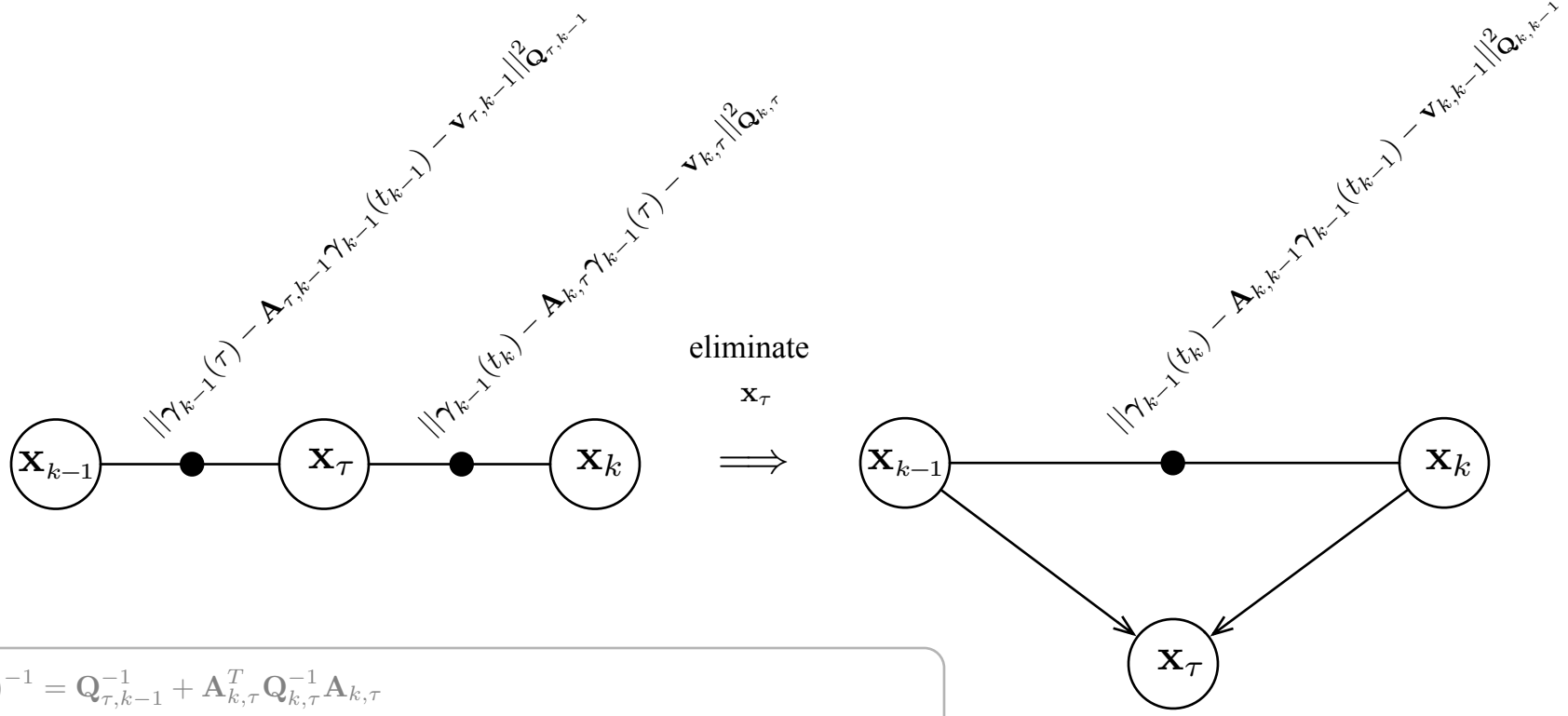

$$\boldsymbol{\Sigma}_{k-1}(\tau)^{-1} = \mathbf{Q}^{-1}_{\tau,k-1} + \mathbf{A}^T_{k,\tau}\mathbf{Q}^{-1}_{k,\tau}\mathbf{A}_{k,\tau}$$

$$\boldsymbol{\Sigma}_{k-1}(\tau)^{-1}\boldsymbol{\mu}_{k-1}(\tau) = \mathbf{Q}^{-1}_{\tau,k-1}\left(\mathbf{A}_{\tau,k-1}\boldsymbol{\gamma}_{k-1}(t_{k-1}) + \mathbf{v}_{\tau,k-1}\right) + \mathbf{A}^T_{k,\tau}\mathbf{Q}^{-1}_{k,\tau}\left(\boldsymbol{\gamma}_{k-1}(t_k) - \mathbf{v}_{k,\tau}\right)$$